# Solar Ti appendix

Sun · Ti · downloadable appendix

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|---|---|
| **Document** | Solar Ti appendix |
| **Subject** | Solar photospheric abundance appendix · Ti |
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Solar titanium: 4.945 ± 0.065

Asplund, Amarsi & Grevesse 2021: 4.97 ± 0.05 · this measurement -0.025 dex

Asplund, Amarsi & Grevesse (2021) give no line recipe we can repeat for this element, so this is our best HARPS product: the Reference Grade analysis with the most complete treatment, then the most lines.

VIS: 4.945 ± 0.065 · HARPS · VIS · Ti I · reference lines · <3D> LTE (Gerber)

## Highlighted Products

### near-UV

**5.106 ± 0.047**

Ti II · reference lines · <3D> LTE (Gerber) · Kitt Peak (1984, molecfit) · Reference Grade

### VIS

**4.940 ± 0.062**

Ti I · reference lines · <3D> LTE (Gerber) · Kitt Peak (Kurucz 2005) · Reference Grade

### red-optical

**5.113 ± 0.094**

Ti I · reference lines · <3D> NLTE (Gerber) · Kitt Peak (Kurucz 2005) · Reference Grade

### NIR

**5.022 ± 0.172**

Ti I · reference lines · <3D> LTE (Gerber) · Kitt Peak (1984, molecfit) · Reference Grade

### H

**4.864 ± 0.122**

Ti I · reference lines · <3D> NLTE (Gerber) · CRIRES+ H (full arm, telluric-corrected) · Reference Grade

### K

**4.932 ± 0.158**

Ti I · reference lines · <3D> NLTE (Gerber) · CRIRES+ · Reference Grade

## Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

| Grade | What it means | This element |
|---|---|---|
| Reference Grade | lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used | 192 products -- Ti I · Asplund et al. 2021 lines; Ti I · reference lines; Ti II · Asplund et al. 2021 lines; Ti II · reference lines |
| Codex Grade | lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep | 138 products -- Ti I · Codex-graded lines; Ti II · Codex-graded lines |
| Deep Grade | the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere | none for the Sun: every line in this grade is excluded as saturated or an outlier (see Excluded lines) |

# Error-bar Forest Plot

## near-UV — Kitt Peak (Kurucz 2005)
## A(Ti) dex; axis 4.68–5.62

| | | |
|---|---|---|
| Ti II · Codex-graded lines · 1D LTE<br>Codex Grade · 5 lines |  | 5.082<br>±0.122 |
| Ti II · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 5 lines |  | 5.082<br>±0.123 |
| Ti II · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 4 lines |  | 5.118<br>±0.111 |
| Ti II · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 5 lines |  | 5.067<br>±0.119 |
| Ti II · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.107<br>±0.101 |
| Ti II · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.202<br>±0.113 |
| Ti II · reference lines · 1D LTE<br>Reference Grade · 153 lines |  | 5.183<br>±0.048 |
| Ti II · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 147 lines |  | 5.184<br>±0.049 |
| Ti II · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 96 lines |  | 5.101<br>±0.041 |
| Ti II · reference lines · 1D LTE (Gerber)<br>Reference Grade · 153 lines |  | 5.203<br>±0.048 |
| Ti II · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 154 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.361<br>±0.048 |
| Ti II · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 135 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.289<br>±0.043 |
| Ti I · Codex-graded lines · 1D LTE<br>Codex Grade · 15 lines |  | 5.206<br>±0.168 |
| Ti I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 15 lines |  | 5.248<br>±0.167 |
| Ti I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 8 lines |  | 5.140<br>±0.142 |
| Ti I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 14 lines |  | 5.204<br>±0.151 |
| Ti I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 14 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.348<br>±0.145 |

## near-UV — Kitt Peak (Kurucz 2005)
## A(Ti) dex; axis 4.68–5.62

| | | |
|---|---|---|
| Ti I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 12 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.455<br>±0.123 |
| Ti I · reference lines · 1D LTE<br>Reference Grade · 34 lines |  | 5.293<br>±0.104 |
| Ti I · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 34 lines |  | 5.318<br>±0.104 |
| Ti I · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 15 lines |  | 5.067<br>±0.084 |
| Ti I · reference lines · 1D LTE (Gerber)<br>Reference Grade · 33 lines |  | 5.303<br>±0.101 |
| Ti I · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 31 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.459<br>±0.108 |
| Ti I · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 26 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.506<br>±0.084 |

## near-UV — Kitt Peak (1984, molecfit)
## A(Ti) dex; axis 4.68–5.62

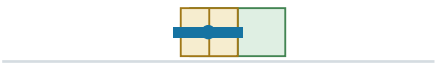
| | | |
|---|---|---|
| Ti II · Codex-graded lines · 1D LTE<br>Codex Grade · 5 lines |  | 4.845<br>±0.125 |
| Ti II · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 5 lines |  | 4.836<br>±0.125 |
| Ti II · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 5 lines |  | 4.948<br>±0.141 |
| Ti II · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 5 lines |  | 4.854<br>±0.117 |
| Ti II · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 5 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.946<br>±0.106 |
| Ti II · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 5 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.035<br>±0.124 |
| Ti II · reference lines · 1D LTE<br>Reference Grade · 153 lines |  | 5.003<br>±0.046 |
| Ti II · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 147 lines |  | 5.012<br>±0.047 |
| Ti II · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 121 lines |  | 4.965<br>±0.040 |
| Ti II · reference lines · 1D LTE (Gerber)<br>Reference Grade · 154 lines |  | 5.016<br>±0.047 |

## near-UV — Kitt Peak (1984, molecfit)
## A(Ti) dex; axis 4.68–5.62

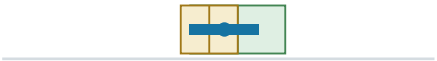
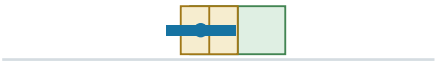
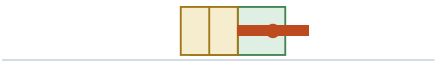
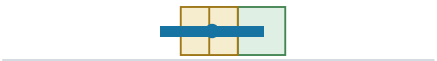
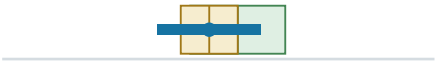
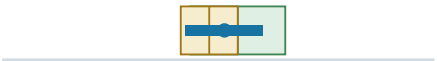
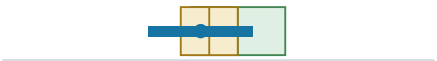
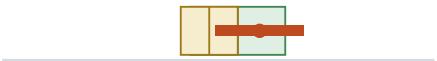
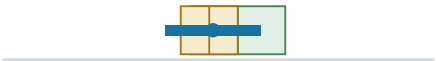
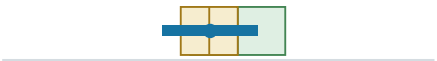
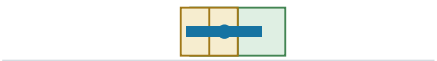
| | | |
|---|---|---|
| Ti II · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 156 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.106<br>±0.047 |
| Ti II · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 142 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.114<br>±0.043 |
| Ti I · Codex-graded lines · 1D LTE<br>Codex Grade · 12 lines |  | 5.047<br>±0.129 |
| Ti I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 12 lines |  | 5.091<br>±0.129 |
| Ti I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 10 lines |  | 5.067<br>±0.136 |
| Ti I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 12 lines |  | 5.055<br>±0.130 |
| Ti I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 14 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.186<br>±0.141 |
| Ti I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 13 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.233<br>±0.130 |
| Ti I · reference lines · 1D LTE<br>Reference Grade · 36 lines |  | 4.963<br>±0.092 |
| Ti I · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 36 lines |  | 4.988<br>±0.092 |
| Ti I · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 28 lines |  | 5.074<br>±0.074 |
| Ti I · reference lines · 1D LTE (Gerber)<br>Reference Grade · 34 lines |  | 5.006<br>±0.076 |
| Ti I · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 35 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.164<br>±0.084 |
| Ti I · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 35 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.280<br>±0.080 |

near-UV axis: 4.68 · 4.92 · 5.15 · 5.38 · 5.62 A(Ti) dex

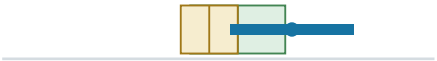
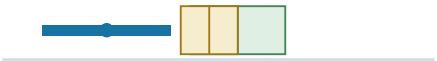
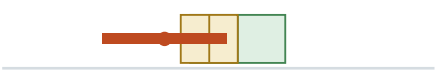
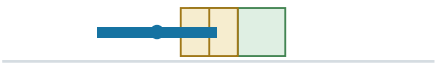
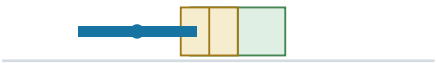
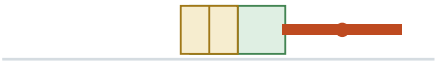
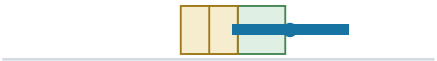
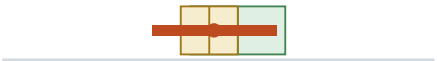
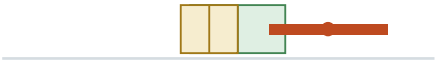
## VIS — HARPS
## A(Ti) dex; axis 4.72–5.18

| | | |
|---|---|---|
| Ti II · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 13 lines |  | 4.939<br>±0.037 |
| Ti II · Asplund et al. 2021 lines · 1D NLTE (Amarsi)<br>Reference Grade · 13 lines |  | 4.938<br>±0.037 |

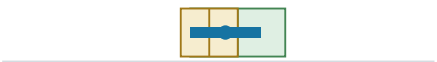
## VIS — HARPS
## A(Ti) dex; axis 4.72–5.18

| | | |
|---|---|---|
| Ti II · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 13 lines |  | 4.956<br>±0.037 |
| Ti II · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 13 lines |  | 4.931<br>±0.036 |
| Ti II · Asplund et al. 2021 lines · <3D> LTE (Gerber)<br>Reference Grade · 13 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.993<br>±0.038 |
| Ti II · Asplund et al. 2021 lines · <3D> NLTE (Gerber)<br>Reference Grade · 13 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.007<br>±0.037 |
| Ti II · Codex-graded lines · 1D LTE<br>Codex Grade · 49 lines |  | 4.943<br>±0.055 |
| Ti II · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 49 lines |  | 4.940<br>±0.055 |
| Ti II · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 47 lines |  | 4.956<br>±0.041 |
| Ti II · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 49 lines |  | 4.931<br>±0.055 |
| Ti II · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 48 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.993<br>±0.046 |
| Ti II · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 48 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.014<br>±0.044 |
| Ti II · reference lines · 1D LTE<br>Reference Grade · 72 lines |  | 4.944<br>±0.050 |
| Ti II · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 72 lines |  | 4.941<br>±0.050 |
| Ti II · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 70 lines |  | 4.956<br>±0.040 |
| Ti II · reference lines · 1D LTE (Gerber)<br>Reference Grade · 72 lines |  | 4.932<br>±0.051 |
| Ti II · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 72 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.995<br>±0.043 |
| Ti II · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 71 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.023<br>±0.042 |
| Ti I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 27 lines |  | 4.819<br>±0.067 |
| Ti I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)<br>Reference Grade · 26 lines |  | 4.845<br>±0.068 |

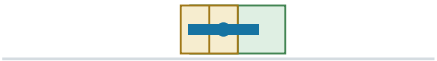
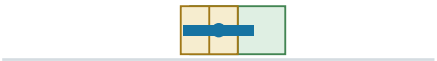
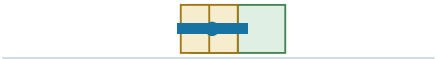
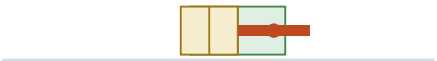
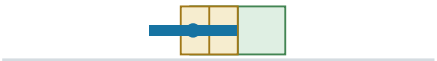
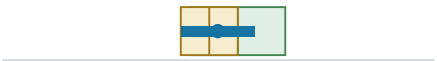
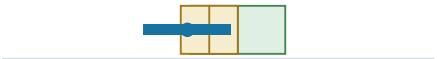
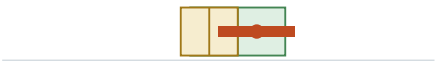
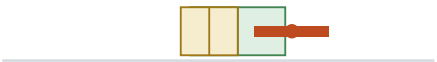
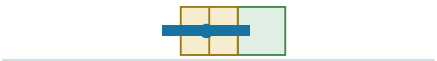
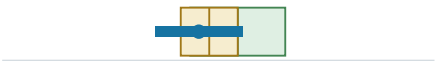
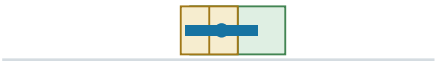
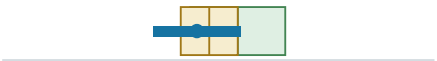
## VIS — HARPS
## A(Ti) dex; axis 4.72–5.18

| | | |
|---|---|---|
| Ti I · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 27 lines |  | 5.027<br>±0.065 |
| Ti I · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 27 lines |  | 4.832<br>±0.067 |
| Ti I · Asplund et al. 2021 lines · <3D> LTE (Gerber)<br>Reference Grade · 27 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.893<br>±0.065 |
| Ti I · Asplund et al. 2021 lines · <3D> NLTE (Gerber)<br>Reference Grade · 27 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.053<br>±0.063 |
| Ti I · Codex-graded lines · 1D LTE<br>Codex Grade · 99 lines |  | 4.848<br>±0.063 |
| Ti I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 97 lines |  | 4.885<br>±0.063 |
| Ti I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 90 lines |  | 5.028<br>±0.062 |
| Ti I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 98 lines |  | 4.864<br>±0.063 |
| Ti I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 104 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.942<br>±0.064 |
| Ti I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 100 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.080<br>±0.063 |
| Ti I · reference lines · 1D LTE<br>Reference Grade · 77 lines |  | 4.838<br>±0.063 |
| Ti I · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 77 lines |  | 4.853<br>±0.063 |
| Ti I · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 73 lines |  | 5.025<br>±0.061 |
| Ti I · reference lines · 1D LTE (Gerber)<br>Reference Grade · 77 lines |  | 4.844<br>±0.063 |
| Ti I · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 80 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.945<br>±0.065 |
| Ti I · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 77 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.065<br>±0.062 |

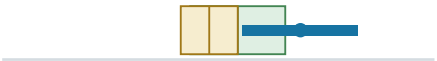
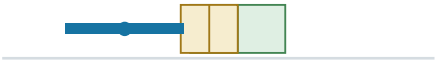
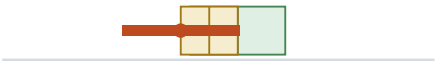
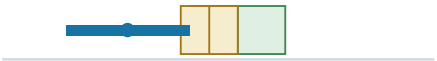
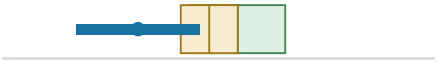
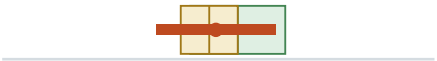
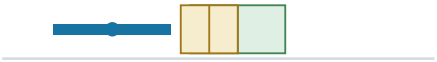
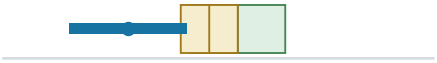
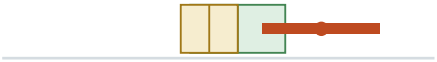
## VIS — Kitt Peak (Kurucz 2005)
## A(Ti) dex; axis 4.72–5.18

| | | |
|---|---|---|
| Ti II · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 14 lines |  | 4.957<br>±0.037 |

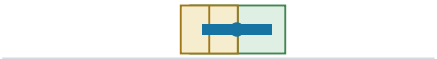
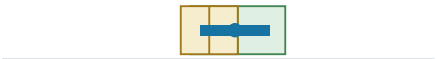
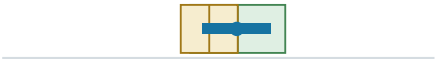
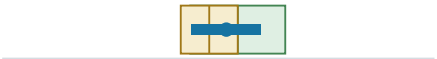
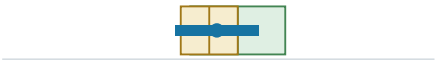
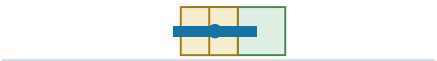
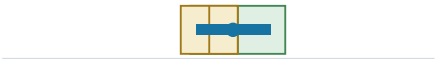
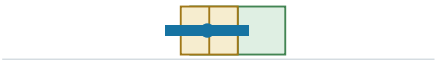
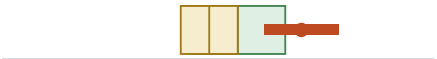
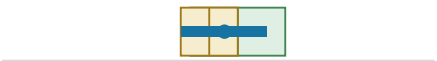
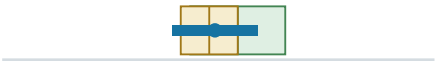
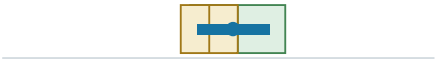
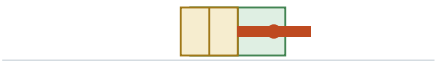
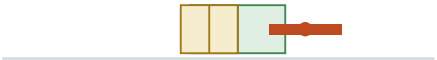
## VIS — Kitt Peak (Kurucz 2005)
## A(Ti) dex; axis 4.72–5.18

| | | |
|---|---|---|
| Ti II · Asplund et al. 2021 lines · 1D NLTE (Amarsi)<br>Reference Grade · 14 lines |  | 4.955<br>±0.037 |
| Ti II · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 14 lines |  | 4.950<br>±0.037 |
| Ti II · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 14 lines |  | 4.943<br>±0.037 |
| Ti II · Asplund et al. 2021 lines · <3D> LTE (Gerber)<br>Reference Grade · 14 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.008<br>±0.037 |
| Ti II · Asplund et al. 2021 lines · <3D> NLTE (Gerber)<br>Reference Grade · 14 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.011<br>±0.037 |
| Ti II · Codex-graded lines · 1D LTE<br>Codex Grade · 49 lines |  | 4.926<br>±0.046 |
| Ti II · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 49 lines |  | 4.923<br>±0.046 |
| Ti II · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 48 lines |  | 4.949<br>±0.039 |
| Ti II · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 49 lines |  | 4.917<br>±0.046 |
| Ti II · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 48 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.990<br>±0.040 |
| Ti II · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 48 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.027<br>±0.039 |
| Ti II · reference lines · 1D LTE<br>Reference Grade · 73 lines |  | 4.937<br>±0.046 |
| Ti II · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 73 lines |  | 4.929<br>±0.046 |
| Ti II · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 71 lines |  | 4.953<br>±0.038 |
| Ti II · reference lines · 1D LTE (Gerber)<br>Reference Grade · 73 lines |  | 4.927<br>±0.046 |
| Ti II · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 73 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.996<br>±0.039 |
| Ti II · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 72 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.030<br>±0.038 |
| Ti I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 27 lines |  | 4.839<br>±0.062 |

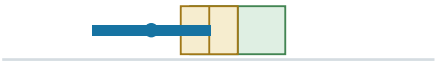
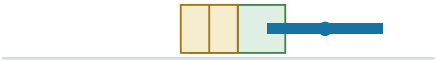
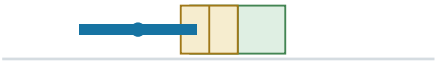
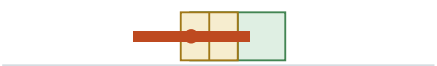
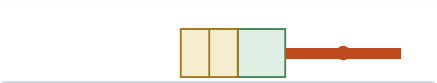
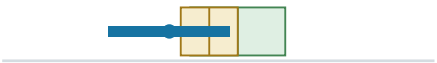
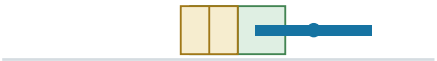
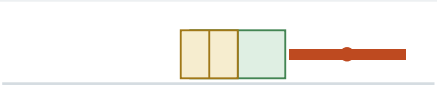
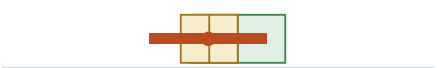
## VIS — Kitt Peak (Kurucz 2005)
## A(Ti) dex; axis 4.72–5.18

| | | |
|---|---|---|
| Ti I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)<br>Reference Grade · 26 lines |  | 4.869<br>±0.063 |
| Ti I · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 27 lines |  | 5.036<br>±0.061 |
| Ti I · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 27 lines |  | 4.851<br>±0.062 |
| Ti I · Asplund et al. 2021 lines · <3D> LTE (Gerber)<br>Reference Grade · 27 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.910<br>±0.062 |
| Ti I · Asplund et al. 2021 lines · <3D> NLTE (Gerber)<br>Reference Grade · 27 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.074<br>±0.060 |
| Ti I · Codex-graded lines · 1D LTE<br>Codex Grade · 104 lines |  | 4.854<br>±0.065 |
| Ti I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 102 lines |  | 4.893<br>±0.065 |
| Ti I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 97 lines |  | 5.043<br>±0.061 |
| Ti I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 104 lines |  | 4.865<br>±0.065 |
| Ti I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 106 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.947<br>±0.063 |
| Ti I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 102 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.085<br>±0.062 |
| Ti I · reference lines · 1D LTE<br>Reference Grade · 78 lines |  | 4.838<br>±0.062 |
| Ti I · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 78 lines |  | 4.855<br>±0.062 |
| Ti I · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 76 lines |  | 5.029<br>±0.061 |
| Ti I · reference lines · 1D LTE (Gerber)<br>Reference Grade · 78 lines |  | 4.854<br>±0.062 |
| Ti I · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 80 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.940<br>±0.062 |
| Ti I · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 78 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.058<br>±0.062 |

## VIS — Kitt Peak (1984, molecfit)
## A(Ti) dex; axis 4.72–5.18

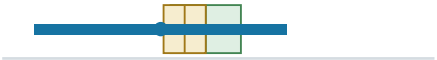
| | | |
|---|---|---|
| Ti II · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 14 lines |  | 4.969<br>±0.037 |
| Ti II · Asplund et al. 2021 lines · 1D NLTE (Amarsi)<br>Reference Grade · 14 lines |  | 4.967<br>±0.037 |
| Ti II · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 14 lines |  | 4.969<br>±0.036 |
| Ti II · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 14 lines |  | 4.958<br>±0.036 |
| Ti II · Asplund et al. 2021 lines · <3D> LTE (Gerber)<br>Reference Grade · 14 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.024<br>±0.037 |
| Ti II · Asplund et al. 2021 lines · <3D> NLTE (Gerber)<br>Reference Grade · 14 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.031<br>±0.036 |
| Ti II · Codex-graded lines · 1D LTE<br>Codex Grade · 49 lines |  | 4.948<br>±0.044 |
| Ti II · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 49 lines |  | 4.946<br>±0.044 |
| Ti II · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 48 lines |  | 4.965<br>±0.039 |
| Ti II · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 49 lines |  | 4.938<br>±0.044 |
| Ti II · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 48 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.995<br>±0.039 |
| Ti II · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 48 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.037<br>±0.039 |
| Ti II · reference lines · 1D LTE<br>Reference Grade · 73 lines |  | 4.956<br>±0.045 |
| Ti II · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 73 lines |  | 4.946<br>±0.045 |
| Ti II · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 71 lines |  | 4.965<br>±0.038 |
| Ti II · reference lines · 1D LTE (Gerber)<br>Reference Grade · 73 lines |  | 4.945<br>±0.045 |
| Ti II · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 73 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.008<br>±0.039 |
| Ti II · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 72 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.041<br>±0.038 |

## VIS — Kitt Peak (1984, molecfit)
## A(Ti) dex; axis 4.72–5.18

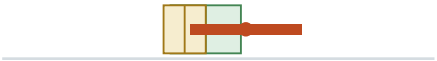
| | | |
|---|---|---|
| Ti I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 27 lines |  | 4.851<br>±0.062 |
| Ti I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)<br>Reference Grade · 26 lines |  | 4.879<br>±0.062 |
| Ti I · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 27 lines |  | 5.062<br>±0.061 |
| Ti I · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 27 lines |  | 4.865<br>±0.062 |
| Ti I · Asplund et al. 2021 lines · <3D> LTE (Gerber)<br>Reference Grade · 27 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.921<br>±0.062 |
| Ti I · Asplund et al. 2021 lines · <3D> NLTE (Gerber)<br>Reference Grade · 27 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.081<br>±0.060 |
| Ti I · Codex-graded lines · 1D LTE<br>Codex Grade · 102 lines |  | 4.859<br>±0.064 |
| Ti I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 100 lines |  | 4.898<br>±0.064 |
| Ti I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 96 lines |  | 5.050<br>±0.061 |
| Ti I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 102 lines |  | 4.872<br>±0.064 |
| Ti I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 105 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.951<br>±0.063 |
| Ti I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 101 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.085<br>±0.061 |
| Ti I · reference lines · 1D LTE<br>Reference Grade · 78 lines |  | 4.853<br>±0.061 |
| Ti I · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 78 lines |  | 4.866<br>±0.061 |
| Ti I · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 75 lines |  | 5.036<br>±0.061 |
| Ti I · reference lines · 1D LTE (Gerber)<br>Reference Grade · 78 lines |  | 4.865<br>±0.061 |
| Ti I · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 80 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.939<br>±0.062 |
| Ti I · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 78 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.075<br>±0.062 |

VIS axis: 4.72 · 4.84 · 4.95 · 5.06 · 5.18 A(Ti) dex

## red-optical — IAG FTS
## A(Ti) dex; axis 4.68–5.30

| | | |
|---|---|---|
| Ti II · Codex-graded lines · 1D LTE<br>Codex Grade · 6 lines |  | 5.018<br>±0.124 |
| Ti II · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 6 lines |  | 5.016<br>±0.126 |
| Ti II · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 5 lines |  | 4.958<br>±0.091 |
| Ti II · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 6 lines |  | 5.006<br>±0.121 |
| Ti II · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.053<br>±0.137 |
| Ti II · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.048<br>±0.131 |
| Ti II · reference lines · 1D LTE<br>Reference Grade · 6 lines |  | 5.018<br>±0.124 |
| Ti II · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 6 lines |  | 5.016<br>±0.126 |
| Ti II · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 5 lines |  | 4.958<br>±0.091 |
| Ti II · reference lines · 1D LTE (Gerber)<br>Reference Grade · 6 lines |  | 5.006<br>±0.121 |
| Ti II · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.053<br>±0.137 |
| Ti II · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.048<br>±0.131 |
| Ti I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 7 lines |  | 4.894<br>±0.180 |
| Ti I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)<br>Reference Grade · 7 lines |  | 4.919<br>±0.173 |
| Ti I · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 7 lines |  | 5.109<br>±0.156 |
| Ti I · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 7 lines |  | 4.906<br>±0.180 |
| Ti I · Asplund et al. 2021 lines · <3D> LTE (Gerber)<br>Reference Grade · 7 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.976<br>±0.154 |
| Ti I · Asplund et al. 2021 lines · <3D> NLTE (Gerber)<br>Reference Grade · 7 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.126<br>±0.132 |

## red-optical — IAG FTS
## A(Ti) dex; axis 4.68–5.30

| | | |
|---|---|---|
| Ti I · Codex-graded lines · 1D LTE<br>Codex Grade · 30 lines |  | 4.918<br>±0.082 |
| Ti I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 29 lines |  | 4.922<br>±0.083 |
| Ti I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 28 lines |  | 5.094<br>±0.077 |
| Ti I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 30 lines |  | 4.931<br>±0.082 |
| Ti I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 30 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.027<br>±0.080 |
| Ti I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 30 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.133<br>±0.077 |
| Ti I · reference lines · 1D LTE<br>Reference Grade · 28 lines |  | 4.918<br>±0.084 |
| Ti I · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 27 lines |  | 4.922<br>±0.085 |
| Ti I · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 26 lines |  | 5.094<br>±0.079 |
| Ti I · reference lines · 1D LTE (Gerber)<br>Reference Grade · 28 lines |  | 4.931<br>±0.084 |
| Ti I · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 28 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.027<br>±0.081 |
| Ti I · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 28 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.133<br>±0.079 |

## red-optical — Kitt Peak (Kurucz 2005)
## A(Ti) dex; axis 4.68–5.30

| | | |
|---|---|---|
| Ti II · Codex-graded lines · 1D LTE<br>Codex Grade · 6 lines |  | 5.025<br>±0.113 |
| Ti II · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 6 lines |  | 5.023<br>±0.114 |
| Ti II · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 5 lines |  | 4.958<br>±0.085 |
| Ti II · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 6 lines |  | 5.013<br>±0.109 |
| Ti II · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.059<br>±0.125 |

## red-optical — Kitt Peak (Kurucz 2005)
## A(Ti) dex; axis 4.68–5.30

| | | |
|---|---|---|
| Ti II · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.059<br>±0.121 |
| Ti II · reference lines · 1D LTE<br>Reference Grade · 6 lines |  | 5.025<br>±0.113 |
| Ti II · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 6 lines |  | 5.023<br>±0.114 |
| Ti II · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 5 lines |  | 4.958<br>±0.085 |
| Ti II · reference lines · 1D LTE (Gerber)<br>Reference Grade · 6 lines |  | 5.013<br>±0.109 |
| Ti II · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.059<br>±0.125 |
| Ti II · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.059<br>±0.121 |
| Ti I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 7 lines |  | 4.879<br>±0.169 |
| Ti I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)<br>Reference Grade · 7 lines |  | 4.904<br>±0.162 |
| Ti I · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 7 lines |  | 5.080<br>±0.139 |
| Ti I · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 7 lines |  | 4.892<br>±0.165 |
| Ti I · Asplund et al. 2021 lines · <3D> LTE (Gerber)<br>Reference Grade · 7 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.962<br>±0.145 |
| Ti I · Asplund et al. 2021 lines · <3D> NLTE (Gerber)<br>Reference Grade · 7 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.108<br>±0.126 |
| Ti I · Codex-graded lines · 1D LTE<br>Codex Grade · 31 lines |  | 4.917<br>±0.086 |
| Ti I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 30 lines |  | 4.909<br>±0.087 |
| Ti I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 29 lines |  | 5.072<br>±0.081 |
| Ti I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 31 lines |  | 4.929<br>±0.084 |
| Ti I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 32 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.012<br>±0.092 |

## red-optical — Kitt Peak (Kurucz 2005)
## A(Ti) dex; axis 4.68–5.30

| | | |
|---|---|---|
| Ti I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 32 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.120<br>±0.090 |
| Ti I · reference lines · 1D LTE<br>Reference Grade · 29 lines |  | 4.915<br>±0.089 |
| Ti I · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 28 lines |  | 4.907<br>±0.090 |
| Ti I · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 27 lines |  | 5.071<br>±0.083 |
| Ti I · reference lines · 1D LTE (Gerber)<br>Reference Grade · 29 lines |  | 4.927<br>±0.087 |
| Ti I · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 30 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.012<br>±0.095 |
| Ti I · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 30 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.113<br>±0.094 |

## red-optical — Kitt Peak (1984, molecfit)
## A(Ti) dex; axis 4.68–5.30

| | | |
|---|---|---|
| Ti II · Codex-graded lines · 1D LTE<br>Codex Grade · 6 lines |  | 5.034<br>±0.117 |
| Ti II · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 6 lines |  | 5.032<br>±0.118 |
| Ti II · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 5 lines |  | 4.986<br>±0.089 |
| Ti II · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 6 lines |  | 5.023<br>±0.114 |
| Ti II · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.069<br>±0.128 |
| Ti II · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.069<br>±0.124 |
| Ti II · reference lines · 1D LTE<br>Reference Grade · 6 lines |  | 5.034<br>±0.117 |
| Ti II · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 6 lines |  | 5.032<br>±0.118 |
| Ti II · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 5 lines |  | 4.986<br>±0.089 |
| Ti II · reference lines · 1D LTE (Gerber)<br>Reference Grade · 6 lines |  | 5.023<br>±0.114 |

## red-optical — Kitt Peak (1984, molecfit)
## A(Ti) dex; axis 4.68–5.30

| | | |
|---|---|---|
| Ti II · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.069<br>±0.128 |
| Ti II · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.069<br>±0.124 |
| Ti I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 7 lines |  | 4.888<br>±0.154 |
| Ti I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)<br>Reference Grade · 7 lines |  | 4.913<br>±0.148 |
| Ti I · Asplund et al. 2021 lines · 1D NLTE (Gerber)<br>Reference Grade · 7 lines |  | 5.111<br>±0.130 |
| Ti I · Asplund et al. 2021 lines · 1D LTE (Gerber)<br>Reference Grade · 7 lines |  | 4.900<br>±0.152 |
| Ti I · Asplund et al. 2021 lines · <3D> LTE (Gerber)<br>Reference Grade · 7 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.970<br>±0.136 |
| Ti I · Asplund et al. 2021 lines · <3D> NLTE (Gerber)<br>Reference Grade · 7 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.124<br>±0.118 |
| Ti I · Codex-graded lines · 1D LTE<br>Codex Grade · 30 lines |  | 4.912<br>±0.081 |
| Ti I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 29 lines |  | 4.913<br>±0.082 |
| Ti I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 28 lines |  | 5.085<br>±0.075 |
| Ti I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 30 lines |  | 4.925<br>±0.080 |
| Ti I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 30 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.012<br>±0.079 |
| Ti I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 30 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.127<br>±0.077 |
| Ti I · reference lines · 1D LTE<br>Reference Grade · 28 lines |  | 4.912<br>±0.083 |
| Ti I · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 27 lines |  | 4.913<br>±0.084 |
| Ti I · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 26 lines |  | 5.085<br>±0.077 |
| Ti I · reference lines · 1D LTE (Gerber)<br>Reference Grade · 28 lines |  | 4.925<br>±0.081 |

## red-optical — Kitt Peak (1984, molecfit)
## A(Ti) dex; axis 4.68–5.30

| | | |
|---|---|---|
| Ti I · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 28 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.012<br>±0.079 |
| Ti I · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 28 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.127<br>±0.078 |

red-optical axis: 4.68 · 4.83 · 4.99 · 5.14 · 5.30 A(Ti) dex

## NIR — IAG FTS
## A(Ti) dex; axis 4.35–5.22

| | | |
|---|---|---|
| Ti II · Codex-graded lines · 1D LTE<br>Codex Grade · 2 lines |  | 4.808<br>±0.084 |
| Ti II · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 2 lines |  | 4.801<br>±0.083 |
| Ti II · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 2 lines |  | 4.806<br>±0.069 |
| Ti II · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 2 lines |  | 4.796<br>±0.083 |
| Ti II · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.832<br>±0.089 |
| Ti II · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.833<br>±0.079 |
| Ti II · reference lines · 1D LTE<br>Reference Grade · 2 lines |  | 4.808<br>±0.084 |
| Ti II · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 2 lines |  | 4.801<br>±0.083 |
| Ti II · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 2 lines |  | 4.806<br>±0.069 |
| Ti II · reference lines · 1D LTE (Gerber)<br>Reference Grade · 2 lines |  | 4.796<br>±0.083 |
| Ti II · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.832<br>±0.089 |
| Ti II · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.833<br>±0.079 |

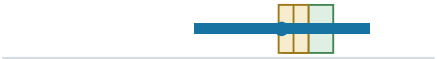
## NIR — Kitt Peak (1984, molecfit)
## A(Ti) dex; axis 4.35–5.22

| | | |
|---|---|---|
| Ti II · Codex-graded lines · 1D LTE<br>Codex Grade · 2 lines |  | 4.841<br>±0.041 |

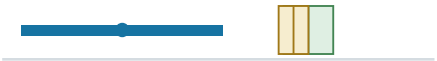
## NIR — Kitt Peak (1984, molecfit)
## A(Ti) dex; axis 4.35–5.22

| | | |
|---|---|---|
| Ti II · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 2 lines |  | 4.834<br>±0.041 |
| Ti II · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 2 lines |  | 4.838<br>±0.045 |
| Ti II · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 2 lines |  | 4.829<br>±0.041 |
| Ti II · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.868<br>±0.037 |
| Ti II · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.870<br>±0.038 |
| Ti II · reference lines · 1D LTE<br>Reference Grade · 2 lines |  | 4.841<br>±0.041 |
| Ti II · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 2 lines |  | 4.834<br>±0.041 |
| Ti II · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 2 lines |  | 4.838<br>±0.045 |
| Ti II · reference lines · 1D LTE (Gerber)<br>Reference Grade · 2 lines |  | 4.829<br>±0.041 |
| Ti II · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.868<br>±0.037 |
| Ti II · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 2 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.870<br>±0.038 |
| Ti I · Codex-graded lines · 1D LTE<br>Codex Grade · 7 lines |  | 4.927<br>±0.176 |
| Ti I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 7 lines |  | 4.917<br>±0.178 |
| Ti I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 6 lines |  | 5.006<br>±0.099 |
| Ti I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 7 lines |  | 4.916<br>±0.177 |
| Ti I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 7 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.022<br>±0.172 |
| Ti I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.065<br>±0.102 |
| Ti I · reference lines · 1D LTE<br>Reference Grade · 7 lines |  | 4.927<br>±0.176 |

## NIR — Kitt Peak (1984, molecfit)
## A(Ti) dex; axis 4.35–5.22

| | | |
|---|---|---|
| Ti I · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 7 lines |  | 4.917<br>±0.178 |
| Ti I · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 6 lines |  | 5.006<br>±0.099 |
| Ti I · reference lines · 1D LTE (Gerber)<br>Reference Grade · 7 lines |  | 4.916<br>±0.177 |
| Ti I · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 7 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.022<br>±0.172 |
| Ti I · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 5.065<br>±0.102 |

## NIR — CRIRES+ J
## A(Ti) dex; axis 4.35–5.22

| | | |
|---|---|---|
| Ti I · Codex-graded lines · 1D LTE<br>Codex Grade · 6 lines |  | 4.595<br>±0.203 |
| Ti I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 6 lines |  | 4.587<br>±0.204 |
| Ti I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 5 lines |  | 4.798<br>±0.137 |
| Ti I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 6 lines |  | 4.614<br>±0.227 |
| Ti I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.722<br>±0.176 |
| Ti I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.840<br>±0.166 |
| Ti I · reference lines · 1D LTE<br>Reference Grade · 6 lines |  | 4.595<br>±0.203 |
| Ti I · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 6 lines |  | 4.587<br>±0.204 |
| Ti I · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 5 lines |  | 4.798<br>±0.137 |
| Ti I · reference lines · 1D LTE (Gerber)<br>Reference Grade · 6 lines |  | 4.614<br>±0.227 |
| Ti I · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.722<br>±0.176 |
| Ti I · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 6 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.840<br>±0.166 |

NIR axis: 4.35 · 4.57 · 4.79 · 5.01 · 5.22 A(Ti) dex

## H — CRIRES+ H
## A(Ti) dex; axis 4.42–5.05

| | | |
|---|---|---|
| Ti I · Codex-graded lines · 1D LTE<br>Codex Grade · 5 lines |  | 4.691<br>±0.241 |
| Ti I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 5 lines |  | 4.718<br>±0.241 |
| Ti I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 4 lines |  | 4.860<br>±0.076 |
| Ti I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 5 lines |  | 4.705<br>±0.240 |
| Ti I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 5 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.766<br>±0.239 |
| Ti I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 4 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.875<br>±0.080 |
| Ti I · reference lines · 1D LTE<br>Reference Grade · 5 lines |  | 4.691<br>±0.241 |
| Ti I · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 5 lines |  | 4.718<br>±0.241 |
| Ti I · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 4 lines |  | 4.860<br>±0.076 |
| Ti I · reference lines · 1D LTE (Gerber)<br>Reference Grade · 5 lines |  | 4.705<br>±0.240 |
| Ti I · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 5 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.766<br>±0.239 |
| Ti I · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 4 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.875<br>±0.080 |

## H — CRIRES+ H (full arm, telluric-corrected)
## A(Ti) dex; axis 4.42–5.05

| | | |
|---|---|---|
| Ti I · Codex-graded lines · 1D LTE<br>Codex Grade · 5 lines |  | 4.669<br>±0.140 |
| Ti I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 5 lines |  | 4.699<br>±0.140 |
| Ti I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 5 lines |  | 4.866<br>±0.134 |
| Ti I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 5 lines |  | 4.691<br>±0.139 |
| Ti I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 5 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.749<br>±0.128 |

## H — CRIRES+ H (full arm, telluric-corrected)
## A(Ti) dex; axis 4.42–5.05

| | | |
|---|---|---|
| Ti I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 5 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.864<br>±0.122 |
| Ti I · reference lines · 1D LTE<br>Reference Grade · 5 lines |  | 4.669<br>±0.140 |
| Ti I · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 5 lines |  | 4.699<br>±0.140 |
| Ti I · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 5 lines |  | 4.866<br>±0.134 |
| Ti I · reference lines · 1D LTE (Gerber)<br>Reference Grade · 5 lines |  | 4.691<br>±0.139 |
| Ti I · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 5 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.749<br>±0.128 |
| Ti I · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 5 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.864<br>±0.122 |

H axis: 4.42 · 4.58 · 4.74 · 4.89 · 5.05 A(Ti) dex

## K — CRIRES+
## A(Ti) dex; axis 4.57–5.12

| | | |
|---|---|---|
| Ti I · Codex-graded lines · 1D LTE<br>Codex Grade · 11 lines |  | 4.748<br>±0.142 |
| Ti I · Codex-graded lines · 1D NLTE (Amarsi)<br>Codex Grade · 11 lines |  | 4.768<br>±0.142 |
| Ti I · Codex-graded lines · 1D NLTE (Gerber)<br>Codex Grade · 9 lines |  | 4.922<br>±0.080 |
| Ti I · Codex-graded lines · 1D LTE (Gerber)<br>Codex Grade · 11 lines |  | 4.764<br>±0.161 |
| Ti I · Codex-graded lines · <3D> LTE (Gerber)<br>Codex Grade · 10 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.853<br>±0.125 |
| Ti I · Codex-graded lines · <3D> NLTE (Gerber)<br>Codex Grade · 11 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.932<br>±0.158 |
| Ti I · reference lines · 1D LTE<br>Reference Grade · 11 lines |  | 4.748<br>±0.142 |
| Ti I · reference lines · 1D NLTE (Amarsi)<br>Reference Grade · 11 lines |  | 4.768<br>±0.142 |
| Ti I · reference lines · 1D NLTE (Gerber)<br>Reference Grade · 9 lines |  | 4.922<br>±0.080 |
| Ti I · reference lines · 1D LTE (Gerber)<br>Reference Grade · 11 lines |  | 4.764<br>±0.161 |

**K — CRIRES+**
**A(Ti) dex; axis 4.57–5.12**

| | | |
|---|---|---|
| Ti I · reference lines · <3D> LTE (Gerber)<br>Reference Grade · 10 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.853<br>±0.125 |
| Ti I · reference lines · <3D> NLTE (Gerber)<br>Reference Grade · 11 lines ·<br>EXPERIMENTAL-NOT-ADOPTED |  | 4.932<br>±0.158 |

K axis: 4.57 · 4.71 · 4.85 · 4.98 · 5.12 A(Ti) dex

**How to read this forest.** Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Each bar is the uncertainty as Asplund et al. 2021 quote it: line-to-line scatter, the 3D-model term and the NLTE term (at least 0.03 dex). For a molecule the NLTE term is replaced by its dependence on the other element, and the 3D term comes from the 3D-to-1D difference Amarsi et al. 2021 publish for the same band. Atomic-data and observational terms are not in the bar, as in Asplund; they stay in the full error budget attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

## How this was measured

Every titanium value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the titanium abundance as the free parameter. The synthesis includes the atomic lines and the molecular opacity in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

| | |
|---|---|
| Band | near-UV |
| What was measured | Ti I · Codex-graded lines; Ti II · Codex-graded lines |
| Solar spectra | Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) |
| Treatments | 1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 4–15 |
| Result A(Ti) | 4.946–5.348<br>(synth-mean3D-LTE-gerber-stagger) |
| Largest uncertainty terms (dex) | fit / line-to-line scatter 0.13, model atmosphere 0.05 |

| | |
|---|---|
| Band | near-UV |
| What was measured | Ti I · reference lines; Ti II · reference lines |
| Solar spectra | Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) |
| Treatments | 1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 15–156 |

| | |
|---|---|
| Band | near-UV |
| What was measured | Ti I · reference lines; Ti II · reference lines |
| Result A(Ti) | 5.106–5.459<br>(synth-mean3D-LTE-gerber-stagger) |
| Largest uncertainty terms (dex) | fit / line-to-line scatter 0.03, model atmosphere 0.02 |

| | |
|---|---|
| Band | VIS |
| What was measured | Ti I · Codex-graded lines; Ti II · Codex-graded lines |
| Solar spectra | HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) |
| Treatments | 1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 47–106 |
| Result A(Ti) | 4.942–4.995<br>(synth-mean3D-LTE-gerber-stagger) |
| Largest uncertainty terms (dex) | model atmosphere 0.05, fit / line-to-line scatter 0.03 |

| | |
|---|---|
| Band | VIS |
| What was measured | Ti I · reference lines; Ti II · reference lines |
| Solar spectra | HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) |
| Treatments | 1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 70–80 |
| Result A(Ti) | 4.939–5.008<br>(synth-mean3D-LTE-gerber-stagger) |
| Largest uncertainty terms (dex) | model atmosphere 0.05, fit / line-to-line scatter 0.02 |

| | |
|---|---|
| Band | VIS |
| What was measured | Ti I · Asplund et al. 2021 lines; Ti II · Asplund et al. 2021 lines |
| Solar spectra | HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) |
| Treatments | 1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 13–27 |
| Result A(Ti) | 5.007–5.081<br>(synth-mean3D-NLTE-gerber-stagger) |
| Largest uncertainty terms (dex) | model atmosphere 0.05, fit / line-to-line scatter 0.02 |

| | |
|---|---|
| Band | red-optical |
| What was measured | Ti I · Codex-graded lines; Ti II · Codex-graded lines |
| Solar spectra | IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) |
| Treatments | 1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 5–32 |
| Result A(Ti) | 5.048–5.133<br>(synth-mean3D-NLTE-gerber-stagger) |
| Largest uncertainty terms (dex) | fit / line-to-line scatter 0.06, model atmosphere 0.06 |

| | |
|---|---|
| Band | red-optical |
| What was measured | Ti I · reference lines; Ti II · reference lines |
| Solar spectra | IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) |

| | |
|---|---|
| Band | red-optical |
| What was measured | Ti I · reference lines; Ti II · reference lines |
| Treatments | 1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 5–30 |
| Result A(Ti) | 5.048–5.133<br>(synth-mean3D-NLTE-gerber-stagger) |
| Largest uncertainty terms (dex) | fit / line-to-line scatter 0.07, model atmosphere 0.06 |

| | |
|---|---|
| Band | red-optical |
| What was measured | Ti I · Asplund et al. 2021 lines |
| Solar spectra | IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) |
| Treatments | 1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 7 |
| Result A(Ti) | 5.108–5.126<br>(synth-mean3D-NLTE-gerber-stagger) |
| Largest uncertainty terms (dex) | fit / line-to-line scatter 0.10, model atmosphere 0.06 |

| | |
|---|---|
| Band | NIR |
| What was measured | Ti I · Codex-graded lines; Ti II · Codex-graded lines |
| Solar spectra | CRIRES+ J, IAG FTS, Kitt Peak (1984, molecfit) |
| Treatments | 1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 2–7 |
| Result A(Ti) | 4.722–5.022<br>(synth-mean3D-LTE-gerber-stagger) |
| Largest uncertainty terms (dex) | fit / line-to-line scatter 0.16, model atmosphere 0.05 |

| | |
|---|---|
| Band | NIR |
| What was measured | Ti I · reference lines; Ti II · reference lines |
| Solar spectra | CRIRES+ J, IAG FTS, Kitt Peak (1984, molecfit) |
| Treatments | 1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 2–7 |
| Result A(Ti) | 4.722–5.022<br>(synth-mean3D-LTE-gerber-stagger) |
| Largest uncertainty terms (dex) | fit / line-to-line scatter 0.16, model atmosphere 0.05 |

| | |
|---|---|
| Band | H |
| What was measured | Ti I · Codex-graded lines |
| Solar spectra | CRIRES+ H, CRIRES+ H (full arm, telluric-corrected) |
| Treatments | 1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 4–5 |
| Result A(Ti) | 4.864–4.875<br>(synth-mean3D-NLTE-gerber-stagger) |
| Largest uncertainty terms (dex) | fit / line-to-line scatter 0.11, model atmosphere 0.05 |

| | |
|---|---|
| Band | H |
| What was measured | Ti I · reference lines |
| Solar spectra | CRIRES+ H, CRIRES+ H (full arm, telluric-corrected) |
| Treatments | 1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 4–5 |
| Result A(Ti) | 4.864–4.875<br>(synth-mean3D-NLTE-gerber-stagger) |
| Largest uncertainty terms (dex) | fit / line-to-line scatter 0.11, model atmosphere 0.05 |

| | |
|---|---|
| Band | K |
| What was measured | Ti I · Codex-graded lines |
| Solar spectra | CRIRES+ |
| Treatments | 1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 9–11 |
| Result A(Ti) | 4.932<br>(synth-mean3D-NLTE-gerber-stagger) |
| Largest uncertainty terms (dex) | fit / line-to-line scatter 0.15, model atmosphere 0.05 |

| | |
|---|---|
| Band | K |
| What was measured | Ti I · reference lines |
| Solar spectra | CRIRES+ |
| Treatments | 1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger |
| Lines | 9–11 |
| Result A(Ti) | 4.932<br>(synth-mean3D-NLTE-gerber-stagger) |
| Largest uncertainty terms (dex) | fit / line-to-line scatter 0.15, model atmosphere 0.05 |

## Solar titanium context

Titanium is measured from neutral Ti I and ionized Ti II lines from the near-ultraviolet to the near-infrared. The reference lines are the Ti I and Ti II lines Asplund et al. 2021 used, with the oscillator strengths they adopted; a wider set of lines with laboratory oscillator strengths and the Codex's own line selection are shown as separate products. Ti I departs from LTE, so each spectrum also carries non-LTE products. No titanium line in the H or K band has a published oscillator-strength uncertainty, so those bands are gaps.

Published line sets · gf uncertainty model · Methodology

## Download the evidence

Download PDF — Solar Ti appendix

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

## Reproducibility

Generated from the pinned science snapshot. Source commit 8efedb8e5221f931d247c2b7b3b1af2cee80decf
Feed Ti.json v1.343 · updated 2026-11-07T11:24:59Z
Generated 2026-11-07T04:26:11-07:00 (source commit timestamp)
Feed SHA-256 b548819f016648d78d2d1fe6ec490245a900125035f8e92048fbc6ad23005f59
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

## References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

**Solar abundance and indicator analyses.** Solar Ti comparison literature; not an adopted Codex abundance. Sources: [1], [2].

**Atomic data and gf authorities.** Ti II oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [3].

**Model and grid sources.** Feed treatment: 1D-LTE synthesis (Turbospectrum); no NLTE/3D correction claimed. Sources: [4].

**Model and grid sources.** Feed treatment: 1D NLTE, per-line correction from the Mallinson et al. 2024 departure grid via PySME (data/nlte_grids/Ti_Mallinson2024_PySME.csv). Sources: [5].

**Model and grid sources.** Feed treatment: ENGINE-B-NLTE; Turbospectrum on the registered Gerber deck. The product retains its own atmosphere and ion identity. Sources: [4].

**Model and grid sources.** Feed treatment: synth-1D-LTE-gerber; Turbospectrum on the registered Gerber deck. The product retains its own atmosphere and ion identity. Sources: [4].

**Model and grid sources.** Feed treatment: synth-mean3D-LTE-gerber-stagger; Turbospectrum on the registered Gerber deck with the horizontally averaged 3D (<3D>) Stagger solar model; not a full 3D calculation. Sources: [4], [6].

**Model and grid sources.** Feed treatment: synth-mean3D-NLTE-gerber-stagger; Turbospectrum on the registered Gerber deck with the horizontally averaged 3D (<3D>) Stagger solar model; not a full 3D calculation. Sources: [4], [6].

**Solar abundance and indicator analyses.** Published line set TI_AGSS21 (the line selection). Sources: [7].

**Atomic data and gf authorities.** Ti I oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [9].

Product-to-source index

NIR · IAG FTS · Ti II · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Ti-products-7c65d6ac7e1e

NIR · IAG FTS · Ti II · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-44b0ca4d0c54

NIR · IAG FTS · Ti II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-1365c96fd02e

NIR · IAG FTS · Ti II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-9cfccbb6cc8f

NIR · IAG FTS · Ti II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-7a6708652ebe

NIR · IAG FTS · Ti II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-cb06c7065b39

NIR · IAG FTS · Ti II · reference lines · 1D LTE · [1], [2], [3], [4] Ti-products-d73e3cc00fb8

NIR · IAG FTS · Ti II · reference lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-077412e3635a

NIR · IAG FTS · Ti II · reference lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-d76f9cfc03a1

NIR · IAG FTS · Ti II · reference lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-483ff025867e

NIR · IAG FTS · Ti II · reference lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-a03f7efb9a31

NIR · IAG FTS · Ti II · reference lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-0e4842d1d04a

NIR · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Ti-products-22a10489f10e

NIR · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-d1354d100be2

NIR · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-a58a333f9167

NIR · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-10627d0beff9

NIR · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-7f36ebf8ef14

NIR · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-54ba69e7d346

NIR · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D LTE · [1], [2], [3], [4] Ti-products-172770c9cf0f

NIR · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-a9144b4d1808

NIR · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-84356bee3afc

NIR · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-33d93daf5341

NIR · Kitt Peak (1984, molecfit) · Ti II · reference lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-958b9fc50c23

NIR · Kitt Peak (1984, molecfit) · Ti II · reference lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-d1f0b6b8d796

VIS · HARPS · Ti II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [7], [3], [4] Ti-products-8899893d2df3

VIS · HARPS · Ti II · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [7], [3], [5] Ti-products-4ef33ba76825

VIS · HARPS · Ti II · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [7], [3], [4] Ti-products-8a6dee5bdc20

VIS · HARPS · Ti II · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [7], [3], [4] Ti-products-9efc551022ee

VIS · HARPS · Ti II · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [7], [3], [4], [6] Ti-products-be051db065f2

VIS · HARPS · Ti II · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [7], [3], [4], [6] Ti-products-14e6b895486c

VIS · HARPS · Ti II · Codex-graded lines · 1D LTE · [1], [2], [3], [8], [4] Ti-products-40a2ff4873f5

VIS · HARPS · Ti II · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [8], [5] Ti-products-0e045b0531b6

VIS · HARPS · Ti II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [8], [4] Ti-products-23d6d4e5035a

VIS · HARPS · Ti II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [8], [4] Ti-products-1c2f938ff19f

VIS · HARPS · Ti II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [3], [8], [4], [6] Ti-products-4bf40df82bcb

VIS · HARPS · Ti II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [3], [8], [4], [6] Ti-products-64130f78edd6

VIS · HARPS · Ti II · reference lines · 1D LTE · [1], [2], [3], [4] Ti-products-0fd1d2acbdc9

VIS · HARPS · Ti II · reference lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-e2646e21f88f

VIS · HARPS · Ti II · reference lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-878de7850327

VIS · HARPS · Ti II · reference lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-a27bb7694c57

VIS · HARPS · Ti II · reference lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-2a0bf00aa87c

VIS · HARPS · Ti II · reference lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-ed779ef895c0

VIS · Kitt Peak (Kurucz 2005) · Ti II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [7], [3], [4] Ti-products-fcfe7475efe3

VIS · Kitt Peak (Kurucz 2005) · Ti II · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [7], [3], [5] Ti-products-0255aa5e70bd

VIS · Kitt Peak (Kurucz 2005) · Ti II · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [7], [3], [4] Ti-products-1233259b48c7

VIS · Kitt Peak (Kurucz 2005) · Ti II · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [7], [3], [4] Ti-products-b5da36a9c162

VIS · Kitt Peak (Kurucz 2005) · Ti II · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [7], [3], [4], [6] Ti-products-71825b39393a

VIS · Kitt Peak (Kurucz 2005) · Ti II · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [7], [3], [4], [6] Ti-products-ca1d61fb34ba

VIS · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D LTE · [1], [2], [3], [8], [4] Ti-products-345d69c8942f

VIS · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [8], [5] Ti-products-53205a6030e0

VIS · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [8], [4] Ti-products-71ed4a6a87e9

VIS · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [8], [4] Ti-products-710ad4a000ba

VIS · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [3], [8], [4], [6] Ti-products-b8e9ca872113

VIS · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [3], [8], [4], [6] Ti-products-09225b61c6fd

VIS · Kitt Peak (Kurucz 2005) · Ti II · reference lines · 1D LTE · [1], [2], [3], [4] Ti-products-27217553b023

VIS · Kitt Peak (Kurucz 2005) · Ti II · reference lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-6a57cd481403

VIS · Kitt Peak (Kurucz 2005) · Ti II · reference lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-3c81ad7ac059

VIS · Kitt Peak (Kurucz 2005) · Ti II · reference lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-b04da3eed446

VIS · Kitt Peak (Kurucz 2005) · Ti II · reference lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-8ab16554cc53

VIS · Kitt Peak (Kurucz 2005) · Ti II · reference lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-eee78383a037

VIS · Kitt Peak (1984, molecfit) · Ti II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [7], [3], [4] Ti-products-809f567843e4

VIS · Kitt Peak (1984, molecfit) · Ti II · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [7], [3], [5] Ti-products-c1b7bcf77fc2

VIS · Kitt Peak (1984, molecfit) · Ti II · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [7], [3], [4] Ti-products-3fb796d44fc7

VIS · Kitt Peak (1984, molecfit) · Ti II · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [7], [3], [4] Ti-products-cf19ec84c534

VIS · Kitt Peak (1984, molecfit) · Ti II · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [7], [3], [4], [6] Ti-products-9d34a92fa771

VIS · Kitt Peak (1984, molecfit) · Ti II · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [7], [3], [4], [6] Ti-products-cee249f5ece0

VIS · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D LTE · [1], [2], [3], [8], [4] Ti-products-c8e6a4e1d084

VIS · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [8], [5] Ti-products-fd905691a44b

VIS · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [8], [4] Ti-products-d48e23581bb2

VIS · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [8], [4] Ti-products-c4da97805634

VIS · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [3], [8], [4], [6] Ti-products-17468c9e5c60

VIS · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [3], [8], [4], [6] Ti-products-c698ed25624e

VIS · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D LTE · [1], [2], [3], [4] Ti-products-7b170fae05f5

VIS · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-eae0967309c8

VIS · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-893394da667a

VIS · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-d51f3bf8014b

VIS · Kitt Peak (1984, molecfit) · Ti II · reference lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-2607900efb7d

VIS · Kitt Peak (1984, molecfit) · Ti II · reference lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-db53f2a0e0c1

near-UV · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Ti-products-099a8856381a

near-UV · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-89bc7ac956bf

near-UV · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-cbd0c69fc2e5

near-UV · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-48817886d33c

near-UV · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-53b6201add71

near-UV · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-f0a094c09798

near-UV · Kitt Peak (Kurucz 2005) · Ti II · reference lines · 1D LTE · [1], [2], [3], [4] Ti-products-673487b1e526

near-UV · Kitt Peak (Kurucz 2005) · Ti II · reference lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-56850348a8ce

near-UV · Kitt Peak (Kurucz 2005) · Ti II · reference lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-dddda6937a4d

near-UV · Kitt Peak (Kurucz 2005) · Ti II · reference lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-c183e3803705

near-UV · Kitt Peak (Kurucz 2005) · Ti II · reference lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-28a7c008c055

near-UV · Kitt Peak (Kurucz 2005) · Ti II · reference lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-aae29a87a1a6

near-UV · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Ti-products-71e27858a2cd

near-UV · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-327bf4cc97a2

near-UV · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-6b278c35888e

near-UV · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-7667b528aa18

near-UV · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-1bede3723ed1

near-UV · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-997e6735dcfa

near-UV · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D LTE · [1], [2], [3], [4] Ti-products-f71002b0523d

near-UV · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-d6a47460ac4e

near-UV · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-9599ffc516e5

near-UV · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-4c4f64d337b1

near-UV · Kitt Peak (1984, molecfit) · Ti II · reference lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-6e5d1155acaf

near-UV · Kitt Peak (1984, molecfit) · Ti II · reference lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-e0ec5eba4607

red-optical · IAG FTS · Ti II · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Ti-products-3e353c565cb8

red-optical · IAG FTS · Ti II · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-70cb0201223f

red-optical · IAG FTS · Ti II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-30b50853129e

red-optical · IAG FTS · Ti II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-df272c5910b2

red-optical · IAG FTS · Ti II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-6c840e2bffca

red-optical · IAG FTS · Ti II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-6fb33d9d5a44

red-optical · IAG FTS · Ti II · reference lines · 1D LTE · [1], [2], [3], [4] Ti-products-bac990455ae3

red-optical · IAG FTS · Ti II · reference lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-2d492768988c

red-optical · IAG FTS · Ti II · reference lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-a026294324fa

red-optical · IAG FTS · Ti II · reference lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-a13da17f4def

red-optical · IAG FTS · Ti II · reference lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-41664373fd73

red-optical · IAG FTS · Ti II · reference lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-545632b99738

red-optical · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Ti-products-d6b3161d26a4

red-optical · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-d96147462172

red-optical · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-05d53ac9404d

red-optical · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-7156ad18391c

red-optical · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-285dd3451742

red-optical · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-355a2510e24e

red-optical · Kitt Peak (Kurucz 2005) · Ti II · reference lines · 1D LTE · [1], [2], [3], [4] Ti-products-4ea9734082e0

red-optical · Kitt Peak (Kurucz 2005) · Ti II · reference lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-e6168291d273

red-optical · Kitt Peak (Kurucz 2005) · Ti II · reference lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-30f30df491c3

red-optical · Kitt Peak (Kurucz 2005) · Ti II · reference lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-3c19808ca9aa

red-optical · Kitt Peak (Kurucz 2005) · Ti II · reference lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-72de71ef567d

red-optical · Kitt Peak (Kurucz 2005) · Ti II · reference lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-162c488c53b5

red-optical · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Ti-products-aacc6bc830bf

red-optical · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-377c69ce7889

red-optical · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-be7439a699e2

red-optical · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-b8b82e4533b0

red-optical · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-e9c4eda0b359

red-optical · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-af2609573153

red-optical · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D LTE · [1], [2], [3], [4] Ti-products-6e35d5b9cdc1

red-optical · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-19e6cc3eaa88

red-optical · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-c223de7313ab

red-optical · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-44b85ba2b6d2

red-optical · Kitt Peak (1984, molecfit) · Ti II · reference lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-a4642f265ac0

red-optical · Kitt Peak (1984, molecfit) · Ti II · reference lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-15eed3926e3a

H · CRIRES+ H · Ti I · Codex-graded lines · 1D LTE · [1], [2], [9], [4] Ti-products-6c7d27968651

H · CRIRES+ H · Ti I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [5] Ti-products-5e7cac644e90

H · CRIRES+ H · Ti I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [9], [4] Ti-products-f4cdbd6af710

H · CRIRES+ H · Ti I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [9], [4] Ti-products-d27fd53799a4

H · CRIRES+ H · Ti I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [9], [4], [6] Ti-products-69e6c2df070c

H · CRIRES+ H · Ti I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [9], [4], [6] Ti-products-48313f3cf66b

H · CRIRES+ H · Ti I · reference lines · 1D LTE · [1], [2], [9], [4] Ti-products-a941d17dd537

H · CRIRES+ H · Ti I · reference lines · 1D NLTE (Amarsi) · [1], [2], [9], [5] Ti-products-e0264700f361

H · CRIRES+ H · Ti I · reference lines · 1D NLTE (Gerber) · [1], [2], [9], [4] Ti-products-333749ac0a0e

H · CRIRES+ H · Ti I · reference lines · 1D LTE (Gerber) · [1], [2], [9], [4] Ti-products-3fcda65c867d

H · CRIRES+ H · Ti I · reference lines · <3D> LTE (Gerber) · [1], [2], [9], [4], [6] Ti-products-40eabc2a73d1

H · CRIRES+ H · Ti I · reference lines · <3D> NLTE (Gerber) · [1], [2], [9], [4], [6] Ti-products-c8132d1ab1cd

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · Codex-graded lines · 1D LTE · [1], [2], [9], [4] Ti-products-f80c402a63be

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [5] Ti-products-02dccdd504fe

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [9], [4] Ti-products-e642100f1077

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [9], [4] Ti-products-7b5355e743e4

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [9], [4], [6] Ti-products-e795ff2b0574

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [9], [4], [6] Ti-products-c8798e88ec5f

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · reference lines · 1D LTE · [1], [2], [9], [4] Ti-products-c77d0a5c90f4

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · reference lines · 1D NLTE (Amarsi) · [1], [2], [9], [5] Ti-products-8df532e260d3

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · reference lines · 1D NLTE (Gerber) · [1], [2], [9], [4] Ti-products-852bfd111d5d

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · reference lines · 1D LTE (Gerber) · [1], [2], [9], [4] Ti-products-4ac3aae52ee3

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · reference lines · <3D> LTE (Gerber) · [1], [2], [9], [4], [6] Ti-products-eef9853714e0

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · reference lines · <3D> NLTE (Gerber) · [1], [2], [9], [4], [6] Ti-products-122c5a981869

K · CRIRES+ · Ti I · Codex-graded lines · 1D LTE · [1], [2], [9], [10], [4] Ti-products-1b0d92bdc785

K · CRIRES+ · Ti I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10], [5] Ti-products-d82f81283c43

K · CRIRES+ · Ti I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [9], [10], [4] Ti-products-06f520f7a07a

K · CRIRES+ · Ti I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [9], [10], [4] Ti-products-04362177ee7b

K · CRIRES+ · Ti I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [9], [10], [4], [6] Ti-products-9540081f59df

K · CRIRES+ · Ti I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [9], [10], [4], [6] Ti-products-33217851a6ee

K · CRIRES+ · Ti I · reference lines · 1D LTE · [1], [2], [9], [10], [4] Ti-products-30ed81c03d88

K · CRIRES+ · Ti I · reference lines · 1D NLTE (Amarsi) · [1], [2], [9], [10], [5] Ti-products-0bd708d01047

K · CRIRES+ · Ti I · reference lines · 1D NLTE (Gerber) · [1], [2], [9], [10], [4] Ti-products-942aba803cb1

K · CRIRES+ · Ti I · reference lines · 1D LTE (Gerber) · [1], [2], [9], [10], [4] Ti-products-3f3b44b8b8ad

K · CRIRES+ · Ti I · reference lines · <3D> LTE (Gerber) · [1], [2], [9], [10], [4], [6] Ti-products-c30cf0a7d6eb

K · CRIRES+ · Ti I · reference lines · <3D> NLTE (Gerber) · [1], [2], [9], [10], [4], [6] Ti-products-67c69701371e

NIR · CRIRES+ J · Ti I · Codex-graded lines · 1D LTE · [1], [2], [10], [4] Ti-products-5ecb885c68a5

NIR · CRIRES+ J · Ti I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [10], [5] Ti-products-61c57ac75c35

NIR · CRIRES+ J · Ti I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [10], [4] Ti-products-e3fa530f51c6

NIR · CRIRES+ J · Ti I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [10], [4] Ti-products-b045a763ba7f

NIR · CRIRES+ J · Ti I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [10], [4], [6] Ti-products-ed112db53aa2

NIR · CRIRES+ J · Ti I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [10], [4], [6] Ti-products-a76810579ac4

NIR · CRIRES+ J · Ti I · reference lines · 1D LTE · [1], [2], [10], [4] Ti-products-67e93afecfd5

NIR · CRIRES+ J · Ti I · reference lines · 1D NLTE (Amarsi) · [1], [2], [10], [5] Ti-products-9ed67a44c882

NIR · CRIRES+ J · Ti I · reference lines · 1D NLTE (Gerber) · [1], [2], [10], [4] Ti-products-fd5b8a8a7f7b

NIR · CRIRES+ J · Ti I · reference lines · 1D LTE (Gerber) · [1], [2], [10], [4] Ti-products-ab012933211f

NIR · CRIRES+ J · Ti I · reference lines · <3D> LTE (Gerber) · [1], [2], [10], [4], [6] Ti-products-865b3c73c214

NIR · CRIRES+ J · Ti I · reference lines · <3D> NLTE (Gerber) · [1], [2], [10], [4], [6] Ti-products-25d111cadbaf

NIR · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · 1D LTE · [1], [2], [9], [10], [4] Ti-products-80519f36b502

NIR · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10], [5] Ti-products-03388a5c7655

NIR · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [10], [4] Ti-products-30c44558556a

NIR · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [9], [10], [4] Ti-products-7d914558f4d0

NIR · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [9], [10], [4], [6] Ti-products-917b9c1ff3ee

NIR · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [10], [4], [6] Ti-products-d0825a7996bc

NIR · Kitt Peak (1984, molecfit) · Ti I · reference lines · 1D LTE · [1], [2], [9], [10], [4] Ti-products-f6a749bb409b

NIR · Kitt Peak (1984, molecfit) · Ti I · reference lines · 1D NLTE (Amarsi) · [1], [2], [9], [10], [5] Ti-products-0cca3a027f81

NIR · Kitt Peak (1984, molecfit) · Ti I · reference lines · 1D NLTE (Gerber) · [1], [2], [10], [4] Ti-products-07b3cca9100d

NIR · Kitt Peak (1984, molecfit) · Ti I · reference lines · 1D LTE (Gerber) · [1], [2], [9], [10], [4] Ti-products-9b19e4010daa

NIR · Kitt Peak (1984, molecfit) · Ti I · reference lines · <3D> LTE (Gerber) · [1], [2], [9], [10], [4], [6] Ti-products-168a22c437cd

NIR · Kitt Peak (1984, molecfit) · Ti I · reference lines · <3D> NLTE (Gerber) · [1], [2], [10], [4], [6] Ti-products-88761131f26f

VIS · HARPS · Ti I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [7], [10], [8], [4] Ti-products-0862b8296512

VIS · HARPS · Ti I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [7], [10], [8], [5] Ti-products-1313f80387b8

VIS · HARPS · Ti I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [7], [10], [8], [4] Ti-products-4a58c624d963

VIS · HARPS · Ti I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [7], [10], [8], [4] Ti-products-ac9b65c076d6

VIS · HARPS · Ti I · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [7], [10], [8], [4], [6] Ti-products-8df9cfede803

VIS · HARPS · Ti I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [7], [10], [8], [4], [6] Ti-products-2ac2463bebb4

VIS · HARPS · Ti I · Codex-graded lines · 1D LTE · [1], [2], [10], [8], [7], [4] Ti-products-a816e4314842

VIS · HARPS · Ti I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [10], [8], [7], [5] Ti-products-46d9d552e533

VIS · HARPS · Ti I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [10], [8], [7], [4] Ti-products-ce1b1402c472

VIS · HARPS · Ti I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [10], [8], [7], [4] Ti-products-ddea97a5e607

VIS · HARPS · Ti I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [10], [8], [7], [4], [6] Ti-products-0288216fc3e7

VIS · HARPS · Ti I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [10], [8], [7], [4], [6] Ti-products-e45736039ceb

VIS · HARPS · Ti I · reference lines · 1D LTE · [1], [2], [10], [9], [4] Ti-products-69d7db25cc24

VIS · HARPS · Ti I · reference lines · 1D NLTE (Amarsi) · [1], [2], [10], [9], [5] Ti-products-1716fd5e5767

VIS · HARPS · Ti I · reference lines · 1D NLTE (Gerber) · [1], [2], [10], [9], [4] Ti-products-864b89fe6a4b

VIS · HARPS · Ti I · reference lines · 1D LTE (Gerber) · [1], [2], [10], [9], [4] Ti-products-75884104f5fb

VIS · HARPS · Ti I · reference lines · <3D> LTE (Gerber) · [1], [2], [10], [9], [4], [6] Ti-products-3d4b572f9848

VIS · HARPS · Ti I · reference lines · <3D> NLTE (Gerber) · [1], [2], [10], [9], [4], [6] Ti-products-1602316590d0

VIS · Kitt Peak (Kurucz 2005) · Ti I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [7], [10], [8], [4] Ti-products-f4fdfcc9c41b

VIS · Kitt Peak (Kurucz 2005) · Ti I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [7], [10], [8], [5] Ti-products-60085323bae2

VIS · Kitt Peak (Kurucz 2005) · Ti I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [7], [10], [8], [4] Ti-products-04c83bb3b37e

VIS · Kitt Peak (Kurucz 2005) · Ti I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [7], [10], [8], [4] Ti-products-0210406ba8ed

VIS · Kitt Peak (Kurucz 2005) · Ti I · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [7], [10], [8], [4], [6] Ti-products-bc2a8979eec3

VIS · Kitt Peak (Kurucz 2005) · Ti I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [7], [10], [8], [4], [6] Ti-products-7ad5fb58dbf6

VIS · Kitt Peak (Kurucz 2005) · Ti I · Codex-graded lines · 1D LTE · [1], [2], [10], [8], [7], [4] Ti-products-ab9987859145

VIS · Kitt Peak (Kurucz 2005) · Ti I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [10], [8], [7], [5] Ti-products-88b5196b2ac0

VIS · Kitt Peak (Kurucz 2005) · Ti I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [10], [8], [7], [4] Ti-products-20c54ac3e35a

VIS · Kitt Peak (Kurucz 2005) · Ti I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [10], [8], [7], [4] Ti-products-fb5b831bba6d

VIS · Kitt Peak (Kurucz 2005) · Ti I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [10], [8], [7], [4], [6] Ti-products-6bfb01cf6907

VIS · Kitt Peak (Kurucz 2005) · Ti I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [10], [8], [7], [4], [6] Ti-products-2011f51a225b

VIS · Kitt Peak (Kurucz 2005) · Ti I · reference lines · 1D LTE · [1], [2], [10], [9], [4] Ti-products-0f302b1fd1e2

VIS · Kitt Peak (Kurucz 2005) · Ti I · reference lines · 1D NLTE (Amarsi) · [1], [2], [10], [9], [5] Ti-products-e9c2390f697e

VIS · Kitt Peak (Kurucz 2005) · Ti I · reference lines · 1D NLTE (Gerber) · [1], [2], [10], [9], [4] Ti-products-58cb5090a64a

VIS · Kitt Peak (Kurucz 2005) · Ti I · reference lines · 1D LTE (Gerber) · [1], [2], [10], [9], [4] Ti-products-932daf83934d

VIS · Kitt Peak (Kurucz 2005) · Ti I · reference lines · <3D> LTE (Gerber) · [1], [2], [10], [9], [4], [6] Ti-products-4bb8b035a608

VIS · Kitt Peak (Kurucz 2005) · Ti I · reference lines · <3D> NLTE (Gerber) · [1], [2], [10], [9], [4], [6] Ti-products-5e569e2fa611

VIS · Kitt Peak (1984, molecfit) · Ti I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [7], [10], [8], [4] Ti-products-f4a6074ee74c

VIS · Kitt Peak (1984, molecfit) · Ti I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [7], [10], [8], [5] Ti-products-baac346cb53c

VIS · Kitt Peak (1984, molecfit) · Ti I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [7], [10], [8], [4] Ti-products-935d497c4a38

VIS · Kitt Peak (1984, molecfit) · Ti I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [7], [10], [8], [4] Ti-products-22fd4ee6fc02

VIS · Kitt Peak (1984, molecfit) · Ti I · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [7], [10], [8], [4], [6] Ti-products-a651eaf7a865

VIS · Kitt Peak (1984, molecfit) · Ti I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [7], [10], [8], [4], [6] Ti-products-e01f293db981

VIS · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · 1D LTE · [1], [2], [10], [8], [7], [4] Ti-products-a89ed80677cd

VIS · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [10], [8], [7], [5] Ti-products-bc548399c5d8

VIS · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [10], [8], [7], [4] Ti-products-382561826847

VIS · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [10], [8], [7], [4] Ti-products-60038af04509

VIS · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [10], [8], [7], [4], [6] Ti-products-da3e8303f035

VIS · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [10], [8], [7], [4], [6] Ti-products-e94b3548e672

VIS · Kitt Peak (1984, molecfit) · Ti I · reference lines · 1D LTE · [1], [2], [10], [9], [4] Ti-products-bc1b7a5a9d0d

VIS · Kitt Peak (1984, molecfit) · Ti I · reference lines · 1D NLTE (Amarsi) · [1], [2], [10], [9], [5] Ti-products-31840e0a0581

VIS · Kitt Peak (1984, molecfit) · Ti I · reference lines · 1D NLTE (Gerber) · [1], [2], [10], [9], [4] Ti-products-7f890c7cc2d6

VIS · Kitt Peak (1984, molecfit) · Ti I · reference lines · 1D LTE (Gerber) · [1], [2], [10], [9], [4] Ti-products-ce70ca34fb4f

VIS · Kitt Peak (1984, molecfit) · Ti I · reference lines · <3D> LTE (Gerber) · [1], [2], [10], [9], [4], [6] Ti-products-8b7a59717a22

VIS · Kitt Peak (1984, molecfit) · Ti I · reference lines · <3D> NLTE (Gerber) · [1], [2], [10], [9], [4], [6] Ti-products-e1ff3c967ac9

near-UV · Kitt Peak (Kurucz 2005) · Ti I · Codex-graded lines · 1D LTE · [1], [2], [8], [10], [4] Ti-products-588c3512c973

near-UV · Kitt Peak (Kurucz 2005) · Ti I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [8], [10], [5] Ti-products-003de2adf2df

near-UV · Kitt Peak (Kurucz 2005) · Ti I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [8], [10], [4] Ti-products-7fad6770e493

near-UV · Kitt Peak (Kurucz 2005) · Ti I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [8], [10], [4] Ti-products-2e11cde72910

near-UV · Kitt Peak (Kurucz 2005) · Ti I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [8], [10], [4], [6] Ti-products-1b6f4ffff3a6

near-UV · Kitt Peak (Kurucz 2005) · Ti I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [8], [10], [4], [6] Ti-products-8fd201520b4a

near-UV · Kitt Peak (Kurucz 2005) · Ti I · reference lines · 1D LTE · [1], [2], [10], [4] Ti-products-9dce97496514

near-UV · Kitt Peak (Kurucz 2005) · Ti I · reference lines · 1D NLTE (Amarsi) · [1], [2], [10], [5] Ti-products-6663d8852598

near-UV · Kitt Peak (Kurucz 2005) · Ti I · reference lines · 1D NLTE (Gerber) · [1], [2], [10], [4] Ti-products-ec827eb0a04d

near-UV · Kitt Peak (Kurucz 2005) · Ti I · reference lines · 1D LTE (Gerber) · [1], [2], [10], [4] Ti-products-e01a7980cf52

near-UV · Kitt Peak (Kurucz 2005) · Ti I · reference lines · <3D> LTE (Gerber) · [1], [2], [10], [4], [6] Ti-products-f0d518fbe7f6

near-UV · Kitt Peak (Kurucz 2005) · Ti I · reference lines · <3D> NLTE (Gerber) · [1], [2], [10], [4], [6] Ti-products-d10fab69fa12

near-UV · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · 1D LTE · [1], [2], [8], [10], [4] Ti-products-fb7e33688a00

near-UV · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [8], [10], [5] Ti-products-88968159e37e

near-UV · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [8], [10], [4] Ti-products-b8261b1e81b6

near-UV · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [8], [10], [4] Ti-products-fa031e1d9fef

near-UV · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [8], [10], [4], [6] Ti-products-ce6e94484977

near-UV · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [8], [10], [4], [6] Ti-products-57bab05188fd

near-UV · Kitt Peak (1984, molecfit) · Ti I · reference lines · 1D LTE · [1], [2], [10], [4] Ti-products-e3aedcf45cc8

near-UV · Kitt Peak (1984, molecfit) · Ti I · reference lines · 1D NLTE (Amarsi) · [1], [2], [10], [5] Ti-products-0bd2a273c334

near-UV · Kitt Peak (1984, molecfit) · Ti I · reference lines · 1D NLTE (Gerber) · [1], [2], [10], [4] Ti-products-ff3c9e3d4e98

near-UV · Kitt Peak (1984, molecfit) · Ti I · reference lines · 1D LTE (Gerber) · [1], [2], [10], [4] Ti-products-160b5e6cd92e

near-UV · Kitt Peak (1984, molecfit) · Ti I · reference lines · <3D> LTE (Gerber) · [1], [2], [10], [4], [6] Ti-products-84f570db5df5

near-UV · Kitt Peak (1984, molecfit) · Ti I · reference lines · <3D> NLTE (Gerber) · [1], [2], [10], [4], [6] Ti-products-7bc4506da852

red-optical · IAG FTS · Ti I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [7], [10], [9], [4] Ti-products-75322b03da34

red-optical · IAG FTS · Ti I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [7], [10], [9], [5] Ti-products-d7395d5abcc1

red-optical · IAG FTS · Ti I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [7], [10], [9], [4] Ti-products-377a151d49b7

red-optical · IAG FTS · Ti I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [7], [10], [9], [4] Ti-products-35c2ce784ff9

red-optical · IAG FTS · Ti I · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [7], [10], [9], [4], [6] Ti-products-78f7457c904e

red-optical · IAG FTS · Ti I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [7], [10], [9], [4], [6] Ti-products-b9c4398f7ce1

red-optical · IAG FTS · Ti I · Codex-graded lines · 1D LTE · [1], [2], [10], [9], [8], [7], [4] Ti-products-5939dd351f06

red-optical · IAG FTS · Ti I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [10], [8], [9], [7], [5] Ti-products-fa14749f3f71

red-optical · IAG FTS · Ti I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [10], [9], [8], [7], [4] Ti-products-592f05a5f767

red-optical · IAG FTS · Ti I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [10], [9], [8], [7], [4] Ti-products-0f1c9884ac13

red-optical · IAG FTS · Ti I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [10], [9], [8], [7], [4], [6] Ti-products-3f9d6b43c2ae

red-optical · IAG FTS · Ti I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [10], [9], [8], [7], [4], [6] Ti-products-f5f8392b41d4

red-optical · IAG FTS · Ti I · reference lines · 1D LTE · [1], [2], [10], [9], [4] Ti-products-dbd18c7fef61

red-optical · IAG FTS · Ti I · reference lines · 1D NLTE (Amarsi) · [1], [2], [10], [9], [5] Ti-products-5c867682d684

red-optical · IAG FTS · Ti I · reference lines · 1D NLTE (Gerber) · [1], [2], [10], [9], [4] Ti-products-65372cc960ae

red-optical · IAG FTS · Ti I · reference lines · 1D LTE (Gerber) · [1], [2], [10], [9], [4] Ti-products-7d431fed8834

red-optical · IAG FTS · Ti I · reference lines · <3D> LTE (Gerber) · [1], [2], [10], [9], [4], [6] Ti-products-6bcde3c96219

red-optical · IAG FTS · Ti I · reference lines · <3D> NLTE (Gerber) · [1], [2], [10], [9], [4], [6] Ti-products-ef77ac6907db

red-optical · Kitt Peak (Kurucz 2005) · Ti I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [7], [10], [9], [4] Ti-products-092f02ba9400

red-optical · Kitt Peak (Kurucz 2005) · Ti I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [7], [10], [9], [5] Ti-products-2aa9849ed089

red-optical · Kitt Peak (Kurucz 2005) · Ti I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [7], [10], [9], [4] Ti-products-2d02d65d5ffc

red-optical · Kitt Peak (Kurucz 2005) · Ti I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [7], [10], [9], [4] Ti-products-43a2edeb8201

red-optical · Kitt Peak (Kurucz 2005) · Ti I · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [7], [10], [9], [4], [6] Ti-products-e6fcbcd00b2e

red-optical · Kitt Peak (Kurucz 2005) · Ti I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [7], [10], [9], [4], [6] Ti-products-cb9c231d1dfe

red-optical · Kitt Peak (Kurucz 2005) · Ti I · Codex-graded lines · 1D LTE · [1], [2], [10], [9], [8], [7], [4] Ti-products-08df19597dbe

red-optical · Kitt Peak (Kurucz 2005) · Ti I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [10], [9], [8], [7], [5] Ti-products-cfec9ff0d013

red-optical · Kitt Peak (Kurucz 2005) · Ti I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [10], [9], [8], [7], [4] Ti-products-f1558ebabe70

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red-optical · Kitt Peak (Kurucz 2005) · Ti I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [10], [9], [8], [7], [4], [6] Ti-products-a6babe027d6d

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red-optical · Kitt Peak (Kurucz 2005) · Ti I · reference lines · 1D LTE · [1], [2], [10], [9], [4] Ti-products-a12c8e6b8e69

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NIR · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-quarantine-57f4084d73ac

NIR · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-quarantine-4a36bee0235b

NIR · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-quarantine-6536e32f8669

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Ti-quarantine-8397baad1d2f

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